Huskysort Revision Assessment

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Overview

The revised *Huskysort* paper is a substantially different paper from the version submitted to SIAM ACDA21 in 2020. This document summarizes how the revision holds up against each of the four original reviewer reports, flags a handful of small items still worth tidying up, and recommends publication venues based on current calls for papers and deadlines (checked as of July 31, 2026).

Bottom line: the revision addresses every substantive point raised by all four reviewers, including the two that mattered most — Reviewer 3's question about radix sort, and Reviewer 4's concern about adversarial/poorly-encoded inputs. The paper now reads as a genuine algorithm-engineering contribution rather than an incremental tweak, which directly answers the Program Committee's stated reason for rejection ("we did not find enough algorithmic innovation").

How the Revision Addresses Each Original Review

Original critique	Status
Review 1: missing cache-oblivious / external-memory literature	**Fixed** — Aggarwal & Vitter and Frigo et al. now cited and explicitly related to the paper's array-access framing
Review 1: Figure 4 as an image; no units; screenshots instead of tables	**Fixed** — clean text tables for environment (Tables 2–3), proper units (ms) throughout
Review 1: "42% faster" obscures the real 1.7x speedup	**Fixed** — paper now argues for and consistently uses multiplier framing (e.g. "1.7x") over percentages
Review 1 / Review 3: encoding phase timing not isolated	**Fixed** — encoding cost measured separately, with concrete numbers (Section 5.2)
Review 3: why not radix sort? (central, previously unanswered question)	**Fixed, thoroughly** — full radix-sort variant (Section 3.2), theoretical model (Section 5.2), empirical results across every data type (Section 6.3), digit-width sweep, and a parallel variant (Section 6.4)
Review 3: too much Java code; poor structure; "husky" name unexplained	**Fixed** — pseudocode algorithms replace most source code; Northeastern mascot explanation included; sections better delineated
Review 4: adversarial inputs / encoding failure modes (deepest unresolved concern)	**Fixed, and the strongest addition** — Appendix A.2 constructs two deliberately adversarial cases (collapsed high-order bits; shared string prefixes) and reports genuinely different, honest outcomes for each
"Single run" / variance concern	**Fixed for the new results** — JMH with 99.9% confidence intervals throughout Tables 8–12; original 2020 numbers clearly labeled as historical/original-environment data

On the adversarial-input finding specifically

This is worth calling out on its own, since it's the strongest piece of new material. The paper tests two distinct failure modes rather than one:

- **Collapsed high-order bits** (many keys share the same leading bits): radix sort's timings stay essentially flat across the whole sweep, exactly as its theoretical model predicts, while a naively-implemented quicksort baseline degrades by more than an order of magnitude and eventually crashes with a stack overflow.
- **Shared string prefixes** exceeding the encoding's capture window: here *nothing* helps — once the encoding carries no disambiguating information at all, every Husky-based approach

(radix included) ends up slower than plain system sort, because the entire ordering burden falls on the mandatory cleanup pass.

Reporting both outcomes honestly — one where the new approach is clearly superior, one where it isn't — is exactly the kind of finding that builds credibility with reviewers who already rejected an earlier, less rigorous version of this same paper.

Small Items Still Worth Fixing Before Submission

1. **Leftover ACM Trans. Graph. template boilerplate** in the header/footer ("ACM Trans. Graph. 37, 4, Article 111…") — needs to be replaced with whichever venue's actual template is used for the final submission.
2. **Classic string-sorting literature** (MSD radix sort, three-way radix quicksort for strings, burstsort) is still not cited. The paper's own radix-sort contribution invites this comparison more directly than before; a reviewer in this specific subfield may ask how Huskysort compares to string-specialized radix/MSD sorts.
3. The **Zhang et al. 2016** "quicksort is fastest" claim still rests on an arXiv-only preprint with no other publication venue listed — minor, but worth being aware a reviewer could flag it.

Venue Recommendations

Checked against current calls for papers as of July 31, 2026.

Venue	Dates	Submission deadline	Notes
SIAM ACDA27	Feb 22–24, 2027, Pittsburgh, PA	Mid-September 2026	Same venue/community that rejected the original submission — resubmitting here directly answers the PC's stated rejection reason. Tightest timeline (~6 weeks from today). Detailed CFP (exact page limits) not yet fully published at time of writing.
SEA 2027	June 21–23, 2027, Karlsruhe, Germany	TBD; likely Jan–Feb 2027 based on SEA 2026's pattern	Best pure thematic fit — SEA's charter is explicitly about experimentation and algorithm-engineering rigor, which is exactly what this paper now demonstrates. More runway than ACDA27.
ALENEX	2027 edition already closed (deadline was July 20, 2026)	Next opportunity: ALENEX 2028 (~1 year out)	Excellent thematic fit, same family as SEA, but not available on a near-term timeline.
Journal of Experimental Algorithmics (JEA)	Rolling submission, no fixed deadline	N/A	Worth considering directly rather than only as a fallback — the paper's appendix material (adversarial testing, partially-sorted-array analysis, coding-accuracy tolerance) is substantial, and a journal format avoids having to compress or relocate any of it to fit a conference page limit.

Recommended path

- If the ~6-week timeline to mid-September is realistic given remaining polish work (template swap, the string-sorting-literature addition, final proofreading), **SIAM ACDA27** is the most narratively satisfying option: the same community, directly shown how thoroughly its critique was addressed.

- If more preparation time is preferred, **SEA 2027** is the strongest pure fit and allows several additional months.
- **JEA** remains a solid option at any point, particularly if the appendix material is judged too valuable to compress into a conference page limit.

Prepared by Claude (Anthropic) at the request of R. Hillyard, based on a review of the revised Huskysort manuscript (PDF supplied July 2026) and the four SIAM ACDA21 reviewer reports.